Gabriel Vance
DSC680 – Applied Data Science
Professor Amirfarrokh Iranitalab
February 22, 2026

Project 3, Milestone 1

Topic

Project name: Screen Time and Health Outcomes in Youth. This project will examine whether higher daily screen exposure is associated with worse mental health, sleep quality, attention, and cognitive performance, and whether these associations differ by age group and type of screen use.

Business Problem

Families, schools, and public health teams need evidence they can trust to decide when screen use becomes harmful and which behaviors are most strongly associated with negative outcomes. The core research questions are: Is higher screen time associated with poorer cognitive and mental health outcomes, which screen behaviors are most predictive of risk, and do sleep and physical activity explain part of the relationship between screen use and health?

Datasets

Primary Kaggle sources (CSV-ready):

1. Screen Time Data: Productivity and Attention Span, https://www.kaggle.com/datasets/muhammadalirazazaidi/screen-time-data-

productivity-and-attention-span, expected fields include screen-time duration, productivity proxies, and attention-related variables.

2. Teen Smartphone Usage and Addiction Impact Dataset, https://www.kaggle.com/datasets/vader24/teen-smartphone-usage-and-addiction-impact-dataset, expected fields include usage intensity, addiction indicators, and psychosocial outcomes.
3. Mobile Phone Usage, https://www.kaggle.com/datasets/memoonaqaiser/mobile-phone-usage, expected fields include device use patterns and behavioral indicators.
4. SmartPhone Data, https://www.kaggle.com/datasets/willianoliveiragibin/smartphone-data, expected fields include smartphone usage characteristics at user level.
5. Synthetic Smartphone Usage Dataset, https://www.kaggle.com/datasets/newton62/synthetic-smartphone-usage-dataset, useful for prototyping feature engineering and model testing before final inference.

Methods

The analysis will start with data quality checks, harmonization of screen-time units, and construction of comparable outcome variables across datasets. I will run descriptive statistics, correlation analysis, and stratified comparisons by age and sex. I will then use multivariable regression models (linear and logistic, depending on outcome) with controls for confounders such as age, sex, socioeconomic indicators, sleep, and physical activity. I

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will test non-linear effects of screen time, interaction effects (for example age by screen-time intensity), and sensitivity analyses across datasets. If needed, I will add propensity score matching or weighting to reduce confounding bias in observational comparisons. Results will report effect sizes, uncertainty intervals, and practical significance, not only p-values.

Ethical Considerations

This topic involves sensitive youth health data, so privacy and de-identification are critical. There is a high risk of over-claiming causality from observational data, so conclusions must be framed as associations unless causal assumptions are explicitly justified. Another risk is stigmatizing specific groups (for example heavy users, lower-income groups, or specific genders), so results must be contextualized with structural factors. Dataset licensing and terms of use must be respected, especially for Kaggle sources and any datasets involving minors.

Challenges and Issues

Most screen-time datasets are self-reported, which introduces recall and social-desirability bias. Different datasets define screen time differently (total hours, social media only, gaming only), creating harmonization challenges. Many datasets are cross-sectional, so reverse causality is possible (for example poor mental health may also increase screen

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use). Missing data and inconsistent coding may require imputation and careful sensitivity checks. Some Kaggle datasets may be synthetic or non-representative, so external validation against CDC and census-based sources is necessary.

References

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